

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools : Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Scenario-Based Assessment

Topic 1: Footprinting Tools – Conducting Competitive Intelligence

Q1. (L3 – Apply) – 15 Marks

A cybersecurity consultant has been assigned to perform a passive security assessment for a company named ABC Technologies. The consultant must collect publicly available information without directly interacting with the company's servers.

Task: Apply suitable footprinting tools to identify the company's:

- domain registration details,
- employee email addresses,
- technology stack,
- and publicly exposed subdomains.

Explain how each selected tool helps in conducting competitive intelligence.

Topic 2: DNS Zone Transfers

Q2. (L3 – Apply) – 15 Marks

During an authorized penetration test, a security analyst suspects that the target organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers.

Task: Apply the appropriate DNS command or tool to verify whether zone transfers are permitted. If the transfer succeeds, explain how the retrieved DNS records can be used to identify internal hosts, mail servers, and subdomains during reconnaissance.
Topic 3: Introduction to Social Engineering – Social Engineering Attacks and Countermeasures

Q3. (L3 – Apply) – 20 Marks

Employees of an organization receive an email claiming to be from the IT department, asking them to verify their login credentials through a provided link because of an urgent security update.

Task: Identify the type of social engineering attack and apply suitable countermeasures that employees should follow before responding to the email. Explain how these actions help prevent credential theft.

Q1. Footprinting Tools – Competitive Intelligence

Aim

To collect publicly available information about **ABC Technologies** without directly interacting with its servers.

Since **ABC Technologies** is a placeholder, use the **authorized domain provided by your faculty/lab**. I'll write it as:

example.com

Replace **example.com** with your authorized lab domain.

1. theHarvester – Emails and subdomains

Check installation:

```
theHarvester -h
```

Run a passive search:

```
theHarvester -d example.com -b crtsh
```

Depending on your Kali version, check available sources with:

```
theHarvester -h
```

Save the results:

```
theHarvester -d example.com -b crtsh -f footprinting_report
```

Record:

- Email addresses
- Subdomains
- Hosts
- IP addresses

2. WHOIS – Domain Registration

Install if necessary:

```
sudo apt install whois -y
```

Then:

```
whois example.com
```

Look for information such as:

- Registrar
- Domain creation/expiry information
- Nameservers
- Registration status

Some domains use privacy protection, so personal registrant information may not be available.

3. DNS Information

Use:

```
dig NS example.com
```

Then:

```
dig MX example.com
```

And:

```
dig A example.com
```

These can reveal nameservers, mail servers, and IP addresses.

4. Technology Stack

If your lab provides **WhatWeb**, you can use it against an authorized target:

whatweb https://example.com

It may identify technologies such as:

- Web server
- CMS
- JavaScript frameworks
- Web technologies

For a strictly passive assessment, prefer publicly available sources rather than intrusive scanning.

Q1 Report Table

Tool	Information Obtained	Purpose
theHarvester	Emails, subdomains, hosts	Employee/domain reconnaissance
WHOIS	Registration information	Domain intelligence
dig	DNS, NS, MX, A records	DNS infrastructure mapping
WhatWeb	Web technologies	Technology-stack identification

Competitive Intelligence Analysis

The collected information can help understand the organization's public infrastructure, technology choices, business contacts, and externally visible services. An attacker could potentially use this information to plan more targeted social-engineering or phishing attempts.

Countermeasures

- Minimize unnecessary public information.
- Protect employee email addresses where possible.
- Regularly review public DNS records.
- Remove obsolete subdomains.

- Monitor public exposure of organizational information.
- Use privacy protection for domain-registration information where appropriate.

```
(kali@kali) [~/tesla-osint]
$ theHarvester -d tesla.com -b crtsh -f tesla.osint
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com
[*] Searching CRTsh.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] Hosts found: 327
*.cn.tesla.com
*.de.tesla.com
*.eu.logs.tesla.com
*.gr.tesla.com
*.logs.tesla.com
*.na.logs.tesla.com
*.nv.tesla.com
*.ny.tesla.com
*.palomito.tesla.com
*.powerhub.energy.tesla.com
*.tx.tesla.com
CitiApiPgPProdV6.tesla.com
CitiApiPgPProdV7.tesla.com
CitiApiSSHProdV6.tesla.com
CitiApiSSHProdV7.tesla.com
akamai-apigateway-vehicleextinfofw-prdsvc-st.tesla.com
akamai-apigateway-vehicleextinfofw-stgsvc-st.tesla.com
ams13-gpgw1.tesla.com
apac-cppm.tesla.com
```

```
(kali@kali) [~/tesla-osint]
$ theHarvester -d tesla.com -b crtsh
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com

An exception has occurred: 502, message='Attempt to decode JSON with unexpected mimetype: text/html', url='https://crt.sh/?q=%25.tesla.com&exclude=expired'
[*] Searching CRTsh.

[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.
```

```
(kali@kali) [~]
$ theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

usage: theHarvester [-h] [-d DOMAIN] [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-v] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c] [-f FI
theHarvester is used to gather open source intelligence (OSINT) on a company or domain.

options:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN    Company name or domain to search.
  -l, --limit LIMIT      limit the number of search results, default=500.
  -s, --start START      Start with result number X, default=0.
  -p, --proxies          Use proxies for requests, enter proxies in proxies.yaml.
  -S, --shodan           Use Shodan to query discovered hosts.
  --screenshot SCREENSHOT Take screenshots of resolved domains specify output directory: --screenshot output_directory
  -v, --virtual-host      Verify host name via DNS resolution and search for virtual hosts.
  -e, --dns-server DNS_SERVER DNS server to use for lookup.
  -t, --take-over        Check for takeovers.
  -r, --dns-resolve [DNS_RESOLVE] Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.
  -n, --dns-lookup        Enable DNS server lookup, default False.
  -g, --dns-brute         Perform a DNS brute force on the domain.
  -f, --filename FILENAME Save the results to an XML and JSON file.
  -w, --wordlist WORDLIST Specify a wordlist for API endpoint scanning.
  -a, --api-scan          Scan for API endpoints.
  -q, --quiet            Suppress missing API key warnings.
  -b, --source SOURCE     intelx, leaklookup, netlas, onyphe, otx, pentesttools, projectdiscovery, rapiddns, rocketreach, securityscorecard, securityTrails,
                        urlscan, venacius, virustotal, whoisxml, yahoo, zoomeye
```

Q2. DNS Zone Transfers

This should be done **only against the DNS server provided for your authorized lab.**

Assume:

Domain: lab.local

DNS Server: 192.168.1.20

Replace these with your actual lab values.

Step 1 — Find the nameserver

```
dig NS lab.local
```

Step 2 — Check the SOA record

```
dig SOA lab.local
```

Step 3 — Test AXFR

```
dig AXFR lab.local @192.168.1.20
```

If the transfer succeeds

You may see records such as:

```
lab.local.      IN SOA ...
```

```
lab.local.      IN NS ...
```

```
www.lab.local.  IN A ...
```

```
mail.lab.local. IN A ...
```

```
server.lab.local. IN A ...
```

This demonstrates that the DNS server permits a zone transfer.

If it fails

You may see:

Transfer failed.

or:

status: REFUSED

This indicates that the server is restricting the transfer.

Save the result

```
dig AXFR lab.local @192.168.1.20 > axfr_result.txt
```

Analysis

A successful unauthorized AXFR can expose:

- Internal hostnames
- IP addresses
- Mail servers
- Subdomains
- Nameservers
- Other DNS records

This information can help an attacker build a map of the organization's infrastructure during reconnaissance.

Security Measures

- Restrict AXFR to authorized secondary DNS servers.
- Use TSIG authentication.
- Block unauthorized zone-transfer requests.
- Separate internal and external DNS.
- Monitor DNS logs.
- Regularly audit DNS configuration.

```
(kali@kali) [ ]
$ dig @192.168.1.10 NS .
;; communications error to 192.168.1.10#53: timed out
;; communications error to 192.168.1.10#53: timed out
;; communications error to 192.168.1.10#53: timed out

; <<>> DiG 9.20.15-2-Debian <<>> @192.168.1.10 NS .
; (1 server found)
;; global options: +cmd
;; no servers could be reached
```

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ dig example.com

; <<>> DiG 9.20.15-2-Debian <<>> example.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 40044
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.      300     IN      A      172.66.147.243
example.com.      300     IN      A      104.20.23.154

;; Query time: 280 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Wed Aug 12 23:46:58 EDT 2026
;; MSG SIZE rcvd: 72

(kali@kali)-[~]
$ dig NS example.com

; <<>> DiG 9.20.15-2-Debian <<>> NS example.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 34052
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.      21600   IN      NS      elliot.ns.cloudflare.com.
example.com.      21600   IN      NS      hera.ns.cloudflare.com.

;; Query time: 212 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Wed Aug 12 23:47:20 EDT 2026
;; MSG SIZE rcvd: 95

(kali@kali)-[~]
```

```
Session Actions Edit View Help
(kali@kali)-[~]
$ dig -x 192.168.1.10

; <<>> DiG 9.20.15-2-Debian <<>> -x 192.168.1.10
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 56094
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;10.1.168.192.in-addr.arpa.  IN      PTR

;; Query time: 12 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Wed Aug 12 23:50:10 EDT 2026
;; MSG SIZE rcvd: 54

(kali@kali)-[~]
$ dig @192.168.1.10 NS .
;; communications error to 192.168.1.10#53: timed out
;; communications error to 192.168.1.10#53: timed out
;; communications error to 192.168.1.10#53: timed out

; <<>> DiG 9.20.15-2-Debian <<>> @192.168.1.10 NS .
; (1 server found)
;; global options: +cmd
;; no servers could be reached

(kali@kali)-[~]
$ nmap -Sv -p 53 192.168.1.10
Failed to resolve/decode supposed IPv4 source address "v": Name or service not known
QUITTING!

(kali@kali)-[~]
$ nmap -sV -p 53 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-12 23:52 EDT
Nmap scan report for 192.168.1.10
Host is up (0.00056s latency).

PORT      STATE      SERVICE VERSION
53/tcp    filtered  domain

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.06 seconds

(kali@kali)-[~]
```


Q3. Social Engineering – Phishing Attack

Identify the Attack

The described attack is **phishing**, specifically a **credential-phishing/social-engineering attack**.

The attacker impersonates the IT department and creates a sense of urgency by claiming that the employee must verify their credentials because of a security update.

Attack Flow

Fake IT Email



Urgent Security Message



Employee Clicks Link



Fake Login Page



Credentials Stolen

Warning Signs

Employees should look for:

1. **Urgent language**
Example: "Verify immediately" or "Your account will be disabled."
2. **Suspicious sender address**
The displayed name may say "IT Department" while the actual email address is different.
3. **Suspicious URL**
The link may lead to a fake website.
4. **Credential request**
Legitimate IT departments should not normally request passwords through email links.
5. **Unexpected security notification**
Employees should independently verify unexpected account-security requests.

Countermeasures

1. Verify the Sender

Check the complete email address rather than trusting the display name.

2. Do Not Click Suspicious Links

Instead of clicking the email link, manually open the organization's known official website or use an established internal portal.

3. Never Provide Passwords

Employees should never enter passwords after following an unexpected email link.

4. Verify Through Another Channel

Contact the IT department using a known phone number or official communication channel.

5. Report the Email

Use the organization's **Report Phishing** mechanism or forward the suspicious message to the security team according to company policy.

6. Use MFA

Multi-factor authentication provides additional protection if a password is accidentally exposed.

7. Email Security Controls

Organizations should use:

- Spam/phishing filters
- URL scanning
- Attachment scanning
- SPF
- DKIM
- DMARC

How These Measures Prevent Credential Theft

The attacker depends on the employee trusting the fake message and entering credentials into a fraudulent website. Verifying the sender, avoiding suspicious links, using known official websites, and reporting the email can break this attack chain.

MFA provides another layer of protection because a stolen password alone may not be sufficient to access the account.

Final Conclusion

Footprinting, DNS reconnaissance, and social engineering represent different stages of a potential attack.

Public Information



Reconnaissance



Identify People & Infrastructure



Social Engineering



Potential Credential Theft

Therefore, organizations should minimize unnecessary information exposure, properly secure DNS infrastructure, train employees to recognize phishing, implement strong email-security controls, and use MFA to reduce the impact of compromised credentials.

nano social_engineering_analysis.txt

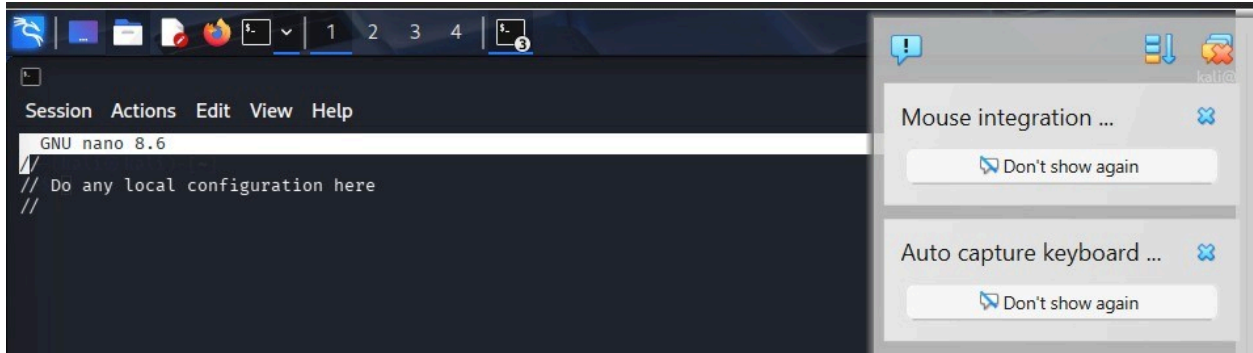
Attack Type: Phishing / Credential Phishing

Warning Signs:

1. Urgent security request
2. Fake IT identity
3. Suspicious link
4. Request for login credentials

Countermeasures:

1. Verify the sender.
2. Do not click suspicious links.
3. Never provide passwords through email.
4. Contact IT using an official channel.
5. Report the email as phishing.
6. Use MFA.



Annexure A

Scenario-Based

Code of Conduct:

I M.NITHIYASHREE-192421101 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.NITHIYASHSREE

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand